
Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

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Abstract

1 A printed claim that the scorecaster breaks online conformal coverage meets its
2 refutation by overlapping authors; this paper joins them and locates the edge
3 of the set that refutation relies on. Conformal PID states long-run coverage for
4 any bounded scorecaster; a corollary keeps it for predictable perturbations of the
5 deployed threshold inside an admissible radius. The claim was acted on; placement
6 and derivation are conceded by name. At the null scorecaster that condition is *tight*.
7 A legal L_1 dead band on the *completed* threshold leaves that set at its radius, and
8 past it miscoverage is 1.000000 against $\alpha = 0.1$. Coverage goes, not only the rate.
9 The edge belongs to the saturator–scorecaster pair, not the readout; elsewhere the
10 condition is sufficient only. Under the equally legal $\hat{q} \equiv +b/2$ the failing band
11 covers; at $\hat{q} \equiv -b/2$ both it and partial adjustment at $w = 0.999$ miscover.

1 Introduction

13 A refereed paper states that a scorecaster breaks Conformal PID’s coverage guarantee. A later one,
14 by the same first and last authors, proves in the same notation that a bounded perturbation of the
15 deployed threshold does not. They share a bibliography, so this is ordinary self-correction, and the
16 joining is what this paper adds.

17 An online conformal method emits a threshold before each observation and moves it. Forecast
18 stability names, scores and prices that movement [Tunc et al., 2013, Godahewa et al., 2025, Van Belle
19 et al., 2023, Pritularga and Kourentzes, 2024, Genov et al., 2026, Van Belle et al., 2026]. Varying
20 temporal structure at fixed level came earlier, in forecast verification, matching predictive marginals
21 on real producers [Gneiting et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]. Coverage
22 and mean length under-describe a deployed interval [Min et al., 2026, Vaze, 2026]. None of it is
23 claimed here.

24 **Claim and refutation, both in print.** Conformal PID [Angelopoulos et al., 2023] adds a *scorecaster*
25 $\hat{q}_{t+1}$ to a saturating error integrator, asking only that it be predictable and lie in $[-b/2, b/2]$: “any
26 function of the past” (Theorem 1). Such a readout is already quantified over. The claim reads
27 “[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical coverage guarantee” [Li and
28 Rodríguez, 2025], stated and acted on in its baselines paragraph, verbatim on refereed printed page
29 18443. It is refuted twice, with proof. Li et al. [2025, Corollary 2] add an arbitrary predictable d_t
30 with $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ to the deployed threshold, note that the result “has exactly the CPID error-
31 integration form”, and conclude $|E_T| = o(T)$. The same generic-slot argument carries AcMCP’s
32 long-run coverage [Wang and Hyndman, 2024]. Four surfaces were queried on 20 August 2026 for
33 both identifiers and “scorecaster”. They are arXiv full text, arXiv abstracts, OpenReview, and the six
34 works Semantic Scholar records citing Li and Rodríguez [2025]. None returns both the claim and the
35 corollary. Hu et al. [2026] still call scorecaster choice “arbitrary” and lacking “principled guidance”
36 while asserting valid coverage: model selection, not validity. **We claim neither the placement nor**
37 **the derivation.**

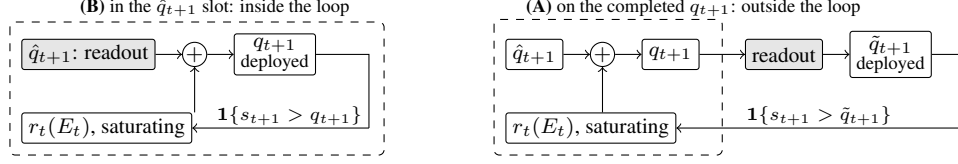


Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop r_t closes. In **(B)** the loop closes on the integrator’s own output, and both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster’s range, inside Corollary 2’s admissible set. In **(A)** it closes on $\tilde{q}_{t+1}$, which it never computed, and the deployed sequence leaves that set.

38 **Where that refutation stops.** Corollary 2 is a *retention* result, silent outside its radius, and the
 39 question it leaves is whether that radius is sharp. Section 3 exhibits a legal perturbation leaving it: the
 40 L_1 dead-band family characterised here on the *completed* threshold, whose edge is the corollary’s
 41 radius at the null scorecaster. Past it coverage goes, not only the rate: the published sufficient condition
 42 is necessary there. Section 4 sets both claims beside four vocabularies for the same movement.

43 2 Setup

44 **The producer.** A forecaster fit before t induces a score s_t , and a threshold q_t is emitted before
 45 y_t is revealed. The deployed set is $C_t = \{y : s_t(x_t, y) \leq q_t\}$ and $\text{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$. The
 46 target is long-run coverage, $T^{-1} \sum_{t \leq T} \text{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID
 47 [Angelopoulos et al., 2023] manipulates the threshold on the score scale,

$$q_{t+1} = \hat{q}_{t+1} + r_t \left(\sum_{i \leq t} (\text{err}_i - \alpha) \right), \quad (1)$$

48 their iteration (5). Its r_t saturates: condition (4) requires $|r_t(x)| \geq b$ with the sign of x once $|x| \geq$
 49 $c h(t)$, for h nonnegative, nondecreasing and sublinear. With $E_t = \sum_{i \leq t} (\text{err}_i - \alpha)$, Proposition 2
 50 gives $|E_T| \leq c h(T) + 1$, read on realised scores.

51 **One scalar, two readouts, two places.** Damping a sequentially updated scalar has two standard
 52 readouts, both prior work. A quadratic cost gives linear partial adjustment $\tilde{q}_{t+1} = (1 - \lambda)\hat{q}_{t+1}^{\text{raw}} + \lambda\tilde{q}_t$
 53 [Gârleanu and Pedersen, 2013], published as post-processing [Godahewa et al., 2025] and applied
 54 to a deployed threshold [Binny and Dixit, 2025, Eq. 13]. A proportional cost gives a no-trade band,
 55 moving only when the target is more than τ away, then pulling back exactly τ [Constantinides,
 56 1986, Davis and Norman, 1990]. Either readout has two positions in (1), drawn in Figure 1; **(A)** is
 57 measured here. *The dead band (L_1) is not a stylistic alternative to the partial-adjustment (L_2) form;*
 58 *it is the family whose failure Section 3 characterises. Neither is safe by construction: L_2 forfeits*
 59 *Proposition 2’s rate while covering, L_1 past τ^* loses coverage, and at the legal $\hat{q} \equiv -b/2$ both lose it.*
 60 *What is tight is the radius, not a form.*

61 **The quantity, and the names it already has.** We report the deployed threshold’s *travel*, $\mathcal{T}_H =$
 62 $\sum_{t=2}^H |q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new. Over a predictive
 63 distribution $\sum |\Delta q|$ is the Multi-Quantile Change [Zanotti, 2026, v3, §3.4.1]; verification calls it
 64 *jumpiness*, a *convergence index*, (*in*)*consistency* [Zsótér et al., 2009, Ehret, 2010, Pappenberger
 65 et al., 2011]; applied control, IACER [Sarhadi, 2025]; dynamic-regret online learning, a path length.
 66 Table 2’s caption names ten, so a fresh one is used here.

67 **Harness.** The harness fixes $\alpha = 0.1$ and $b = 2$. Its saturator $r_t(x) = b \text{clip}(x/(c h(t)), -1, 1)$,
 68 with $c = 1$ and $h(t) = \log(t + 2)$, meets (4) with equality. The scorecaster is null and legal, reducing
 69 (1) to the pure error integration Proposition 2 addresses. Neither choice is neutral: (4) is a lower
 70 bound met exactly here, and Theorem 1 permits scorecasters $b/2$ either side of null. The adversary is
 71 specified, not tie-broken. It plays $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$, $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$: the
 72 $\varepsilon \downarrow 0$ reading of “sets the score at the threshold”, firing the strict indicator. Legality is Theorem 1’s
 73 $[-b/2, b/2]$ on both sides, so $b/2 + b/2 = b$ is exactly tight.

Table 1: Placement A: a one-scalar readout on the *completed* threshold, under one deterministic adversary, on one pass to $T = 10^6$; all eleven arms run are listed. Proposition 2’s bound is 13.2061 at $T = 2 \times 10^5$ and 14.8155 at $T = 10^6$; “ratio” is $\max_t |E_t|$ over it, and “Sat.” the fraction of rounds on which (4) delivers its full $\pm b$. Every figure is the *primary* regime’s, defined in Appendix A, which also gives the $T = 10^4$ excursions and the fitted excursion law.

Readout on the completed threshold	miscov. 2×10^5	miscov. 10^6	$\max_t E_t 10^6$	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	0.100007	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	0.100007	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	0.100007	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	0.100006	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	0.100004	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	0.100006	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	0.100024	53.00	3.58	0.3181	16
running mean	0.099940	0.100010	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	0.100005	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	0.100012	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	1.000000	1.000000	900,000	60,747	1.0000	0.5

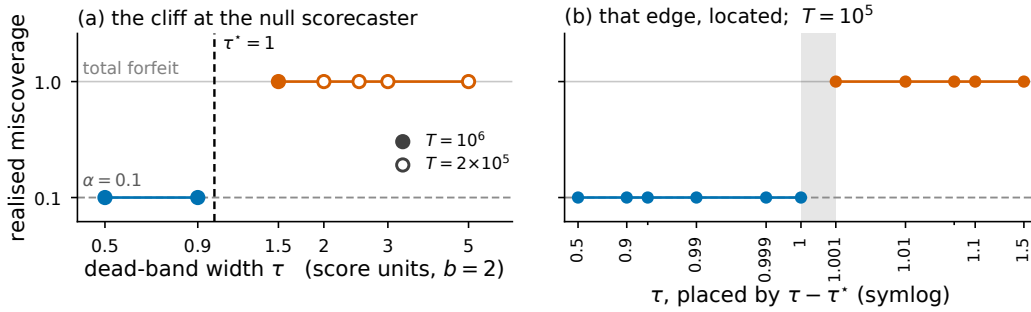


Figure 2: Realised miscoverage against dead-band width τ at the null scorecaster, $\hat{q} \equiv 0$ with a saturator of level b , where $\tau^* = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$ is 1. (a) The full sweep: filled markers run to $T = 10^6$ and the four widest bands, drawn open, to $T = 2 \times 10^5$; blue covers, orange forfeits, and the dashed rule is τ^* . (b) The same setting at $T = 10^5$ over eleven measured widths, placed by $\tau - \tau^*$ on a symlog axis whose linear zone is $|\tau - \tau^*| \leq 10^{-3}$, each tick carrying the τ it marks. The grey band spans the widest covering width to the narrowest failing one; Figure 3 gives the other four $(r_t, \hat{q})$ settings.

74 3 The edge of the admissible set, measured

75 **The retention result is published, and tested from outside.** Li et al. [2025, Corollary 2] prove
76 that a predictable d_t with $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ on the deployed threshold keeps (1) in error-integration
77 form and *retains* $|E_T| \leq c h(T) + 1$: the radius is theirs. A downstream partial adjustment of weight
78 w leaves it; Table 1 prices that departure, and five of its eleven arms exceed the bound at $T = 10^6$.
79 For $w = 0.99$ and 0.999 the excursion is *flat* in T while the bound grows like $\log T$, so the ratio
80 falls with the horizon. At $w = 0.999$ it is 61.08 at $T = 10^4$ and 42.10 at $T = 10^6$. The forfeit is a
81 property of the smoothing *strength*, not Placement A, and it buys something: travel falls from 28,311
82 to 202. The forfeit’s sign is counter-intuitive. A readout on the completed q_{t+1} touches neither r_t nor
83 the integrator’s argument, so it cannot put (4) at risk. It delays a correction already computed, and the
84 accumulator excurses *further*. Smoothing the output does not damp E_t ; it feeds it. That 623.70 turns
85 on the adversary’s ε , not the counting convention.

86 **The L_1 dead band leaves the radius, and the edge is measurable.** A dead band pulls the deployed
87 threshold back by exactly τ when it releases, so “inside Corollary 2” reads $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$. Under
88 saturation the deployed value is pinned τ below the integrator’s ceiling, and its realised supremum
89 is $b - \tau$ to the printed digit at every $\tau \geq 0.9$. The adversary’s best legal play is $b/2$, so coverage
90 is lost when that supremum drops below it, at $\tau^* = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$. All 19 dead-band
91 runs agree with that law (Figure 3); only the null scorecaster’s edge is located (Figure 2), the others
92 one-sided. **Here** $\sup_x r_t(x) = b$ and $\hat{q} \equiv 0$, so $\tau^* = b/2$, one point of the admissible class, and
93 both terms move it. The legal $\hat{q} \equiv +b/2$ carries the $\tau = 1.5$ band inside the bound. The equally
94 legal $\hat{q} \equiv -b/2$ drops the edge to 0, and there $\tau = 0.5$ returns 1.000000, as does partial adjustment

at $w = 0.999$. Condition (4) bounds $\sup_x r_t(x)$ only from below, so raising it costs the theorem nothing; it moves the edge too and shrinks the excursion (Figure 3). Under the unbounded tangent integrator Angelopoulos et al. [2023] report using in all their experiments, τ^* is unbounded and all three widths there cover. The excursion is 20.10 rather than 623.70, both at $T = 10^6$. **The method’s own integrator therefore sits inside the set at every width run here.** Outside the set the transition is a step. Past τ^* the threshold sits permanently below the reachable score range and $|E_t| = (1 - \alpha)t$ grows linearly, 60,747 times the bound at $T = 10^6$. The wider bands $\tau = 2.0, 2.5, 3.0$ and 5.0 fail too at $T = 2 \times 10^5$, so the failing set is (τ^*, ∞) , open at the left. **The endpoint is where the adversary’s specification decides the answer.** At $\tau = \tau^*$ the deployed supremum equals the adversary’s own cap $b/2$, so the clip forces an exact play and the reading settles it: the primary regime covers. Appendix A separates the two readings and prints both. **The failing arms have** Sat. = 1.0000. Condition (4) delivers its full $\pm b$ on every round, and what fails is Proposition 2’s other hypothesis: the threshold closing the loop *is* the integrator’s output.

What that measures is that the published condition is tight. At $\mu_t = 1$, $\hat{q} \equiv 0$ the measured τ^* is exactly Corollary 2’s radius, and crossing it costs coverage rather than the rate. **The claim is that necessity: a tightness result about someone else’s sufficient condition**, smaller and more precise than a forfeit of the rate by post-processing. Elsewhere it stays sufficient only: at $\hat{q} \equiv +b/2$ Corollary 2 permits radius 0 while every width run there covers. That asymmetry makes “tight” a located claim.

The constructive reading, and it is the corollary’s rather than ours. Placement B keeps $\{\hat{q}_t\}$ inside the admissible radius, so Theorem 1 carries it already. What the measurement adds is that the band has an edge, computable before deployment from $\sup_x r_t(x)$ and $\sup_t \hat{q}_t$, and that crossing it is not a graceful degradation.

4 Where this sits

Four vocabularies name one quantity many ways, and the object that settles its validity (the admissible set around the deployed threshold) sits in the first of them. Table 2 in Appendix B names them and what each contributes. **What is conceded, by name.** The placement is prior work: Angelopoulos et al. [2023] state coverage for any admissible integrator and any scorecaster, and run a smoothing-family model in that slot themselves. Dupuy et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too, and this concession matters most. Li et al. [2025, Corollary 2] prove the retention statement in Conformal PID’s own notation, with Wang and Hyndman [2024] an independent second instance. Section 2 concedes the smoother object, the metric arithmetic and both readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path: free-until-the-bound-binds is its defining specification, not a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score. Their one non-standard assumption is that $\sum_{t \leq T} q_t \leq O(\alpha h(T))$ for a sublinear h , supported with a plot rather than derived.

5 Limitations

The evidence is one construction, one adversary, one saturator; and $0.63/(1 - w)$ is specific to the saturator’s *level*, which condition (4) bounds only from below. **The inherited guarantee is weak where it applies.** Under the tangent integrator $h(t) = t/\log t$, the certified gap $(ch(T) + 1)/T$ decays as $O(1/\log T)$ rather than $O(1/T)$. What is forfeited was not a tight certificate. **The dead-band widths are absolute, and the failure is total only against the adversary.** τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75b$. Under i.i.d. scores it returns 0.249376 at $T = 10^6$, an adversary-free confirmation of the mechanism rather than a retreat. The threshold pins at $b - \tau = 0.5$, and uniform scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly. **The harness was rebuilt against numbers already read**, which carries less weight than a rerun done without sight of them. **Placement B is conceded, not tested. One seed, and only the null scorecaster’s edge is bracketed;** the others are one-sided. **What would overturn the claim** is a legal $(r_t, \hat{q})$ pair with $\tau \leq \tau^*$ that loses coverage, or $\tau > \tau^*$ that keeps it; the sweep has neither.

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234 A Supporting measurements

235 **The primary regime.** Every figure in Table 1 is measured under one deterministic adversary. It
 236 plays $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ with $\varepsilon = 10^{-9}$ against the *deployed* threshold, and the indicator is
 237 strict. The specification is not a tie rule, and the distinction is measured rather than argued. Scoring
 238 an exact-threshold play as a *miscoverage* reproduces the primary regime on every cell of Table 1: 44
 239 of 44 identical, all eleven arms at all four horizons. Scoring the same play under the *strict* indicator
 240 does not, and it moves every row, because the adversary’s signature move then covers rather than
 241 misses: partial adjustment at $w = 0.999$ excurses 70.80 in place of 623.70, and the failing dead band
 242 returns 0.000000 in place of 1.000000. The $\varepsilon \downarrow 0$ limit the harness specifies is the first reading. **No**
 243 **row is convention-free with respect to the second**, which is why the adversary is specified rather
 244 than tie-broken. At $\tau = \tau^*$ itself the two readings part: the strict indicator returns 0.000000 where
 245 scoring the exact play as a miscoverage returns 1.000000.

246 **Excursions at $T = 10^4$, and the law across the arms.** Partial adjustment excurses 6.30, 63.00
 247 and 623.70 at $w = 0.9, 0.99$ and 0.999 . The last two are flat in T while Proposition 2’s bound
 248 grows like $\log T$, which is why the ratio falls with the horizon. Across the arms the excursion fits
 249 $\max_t |E_t| \approx \max\{0.5 h(T) + 0.9, 0.63/(1 - w)\}$. This is a fit to the measured arms, not a derived
 250 bound.

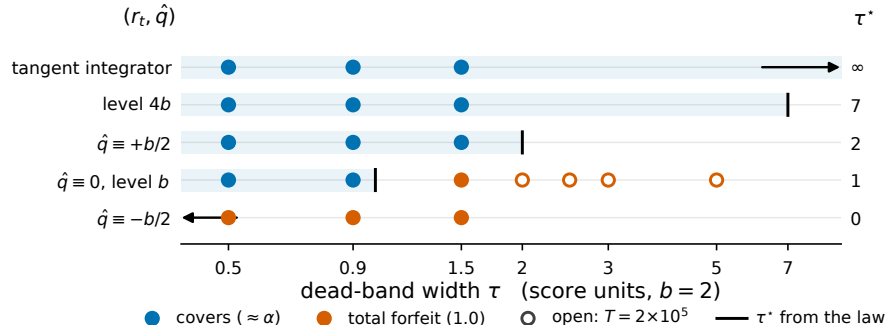


Figure 3: All nineteen dead-band runs, at the five admissible $(r_t, \hat{q})$ settings: rows are settings ordered by τ^* , and the shared horizontal axis is τ . Blue markers cover at $\approx \alpha$ and orange markers return 1.000000; filled markers run to $T = 10^6$ and open markers to $T = 2 \times 10^5$. The rule on each row is that setting’s $\tau^* = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, printed at the right, and the tinted band is $\tau \leq \tau^*$. Two rules fall off the axis and are drawn as arrows: $\tau^* = 0$ at $\hat{q} \equiv -b/2$, and τ^* unbounded under Angelopoulos et al.’s tangent integrator.

251 **Realised miscoverage at the covering settings of Figure 3.** At $\hat{q} \equiv +b/2$ the $\tau = 1.5$ band returns
 252 0.100010 with $\max_t |E_t| = 11.20$, inside the bound. Under Angelopoulos et al.’s tangent integrator
 253 the same width returns 0.100001. Raising the saturator’s level to $4b$ leaves coverage intact and cuts
 254 the $w = 0.999$ excursion from 623.70 to 120.60, which is the first term of τ^* moving the edge rather
 255 than the readout changing behaviour.

256 B The four vocabularies

Table 2: Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness*, *convergence index*, *(in)consistency*; and *switching cost*. The row gives what each contributes.

Online conformal	Forecast stability	Verification	Switching-cost online learning
a distribution-free coverage guarantee for any bounded scorecaster, with a rate [Angelopoulos et al., 2023, Thm 1, Prop. 2], and the admissible set measured here to be tight	long-run- the scored movement functional, the readout map as post-processing [Godaheva et al., 2025, Eq. 3], and a priced decision [Genov et al., 2026, §4.2] and [Van Belle et al., 2026]	the matched-level design on real producers from 2012, with a con- trol <i>stronger</i> than the pair: the whole marginal [Pinson and Gi- rard, 2012, Worsnop et al., 2018]	worst-case theory for the penalised problem: Thm 2, and a Thm 7 trade-off in which sublinear regret is bought by growing θ with T , which grows the ratio [Andrew et al., 2013]